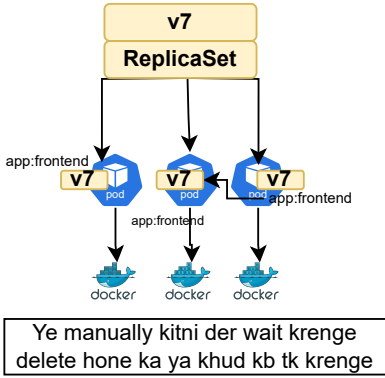
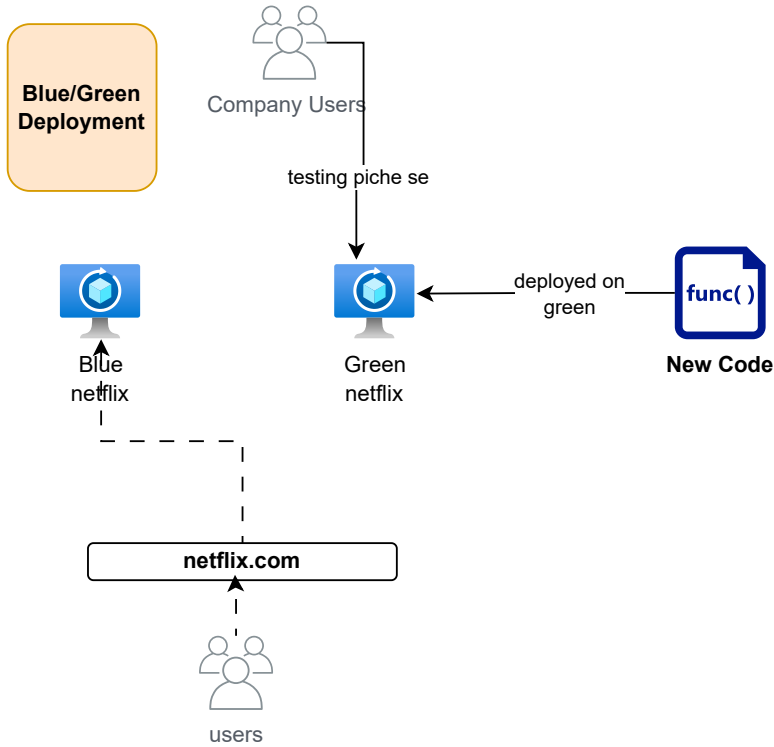
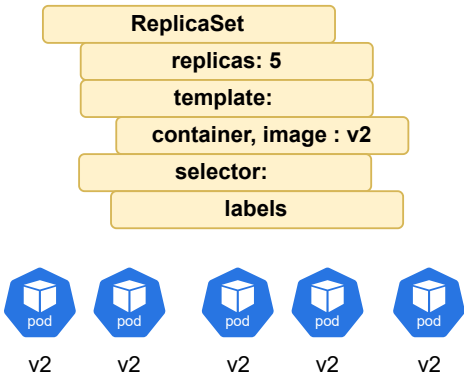


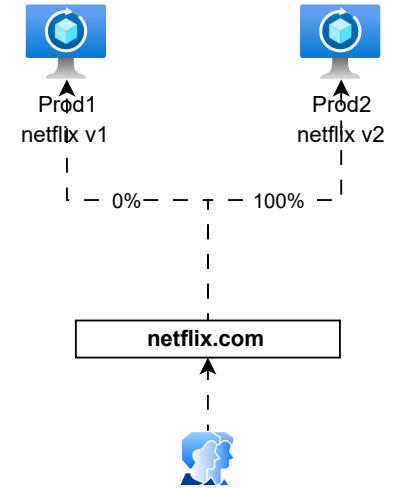


HTTP Error Codes
502 - Bad Gateway
500 - Internal Server Error
501 - Not implemented
503 - Service Unavailable
404 - Not Found
200 - Success
400 - Bad Request
401 - Unauthorised
403- Forbidden

Problems in Replica Set
1. Downtime because no rolling update
2. No Rollback Support as no deployment history maintained
3. Replica Set ensures the desired number of pod replicas are always running, It does not support bluegreen, canary deployments
4. It is not good for CICD automation
5. ReplicaSet in Production is not recommended

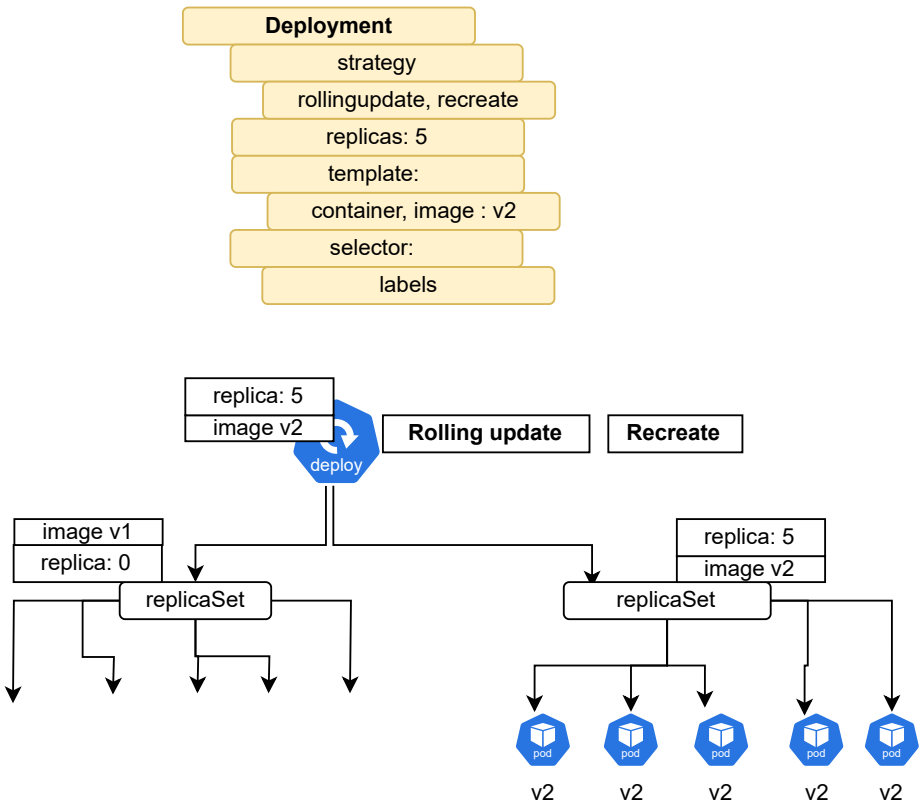


Canary Deployment



Release to a small % of users

Dheere dheere traffic badate jate hai new deployment pr agar sb sahi chلتa hai toh



Developer
Code
MultiStage Docker File
Push on git
Git Flow branching Strategy
Setup agent to build
CI pipeline for docker build
Run Checkmarx, SAST, DAST and other scans
Push Image on ACR
ImagepullSecret inside Cluster
deployment.yaml
service.yaml
Ingress Controllers Setup
ingress rules
secret.yaml, configmap.yaml
secretproviderclass.yaml
hpa.yaml
statefulset.yaml
Gateway API
Service Account.yaml
Role.yaml
ClusterRole.yaml
prometheus/grafana
GitOps ke through CICD
Servicemesh Istio
certificate management using certmanager

Pod agar fail ho gya toh kaise troubleshoot karoge?
Pod Errors ke baare me padna hai - pending, crashloopbackoff, imagepullbackoff, create container config error, runtime container error, OOM killed, evicted, terminating, unknown
Scheduling Errors: Failed Scheduling, Insufficient CPU/Memory, Node Not ready, Node Selector Mismatch, Taint and toleration mismatch
Image Errors: image Pull backoff, Invalid image name, error image pull, invalid image tag
Volume and Storage Errors, Network Errors, Service and Ingress errors, API Server Errors, Authentication and Authorization Errors, deployment errors, stateful set errors, node errors, certificate and TLS errors, Container run time errors